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Uncertainty estimation for multiphenomenology explosion monitoring

Brian J. Williams ,[1] Rick Picard[1] and Dale N. Anderson[2]

[1] *Los Alamos National Laboratory, Statistical Sciences, Los Alamos, NM 87545, USA. E-mail: brianw@lanl.gov*
[2] *Desert Research Institute, Hydrologic Sciences, Las Vegas, NV 89119, USA*

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SUMMARY

We develop and demonstrate a new paradigm for modelling prompt forensics data from potential nuclear explosions of concern. Related scenarios include nuclear terrorism, which may involve low-yield detonations. Traditional modelling appropriate for higher-yield historical nuclear testing is generalized to capture uncertainties in yields (such as when using conversion to obtain 'equivalent' nuclear yields for conventional explosives in low-yield experiments) and to capture variation among source-to-sensor path effects for which no calibration data are available. Special attention is paid to quantifying these sources of uncertainty and to their formal inclusion in comprehensive yield uncertainty—uncertainty that would otherwise be underestimated and potentially lead to mistaken conclusions. For the example scenario considered, two useful stand-alone monitoring phenomenologies are based on geophysical data (from surface effects characteristics and local seismic metrics). By fusing signatures from multiple phenomenologies, the multiphenomenology explosion monitoring framework provides improved yield characterization relative to reliance on single-phenomenology analysis alone, especially when individual sensors have complementary sensitivities to emplacement/environmental conditions.

Key words: Inverse theory; Probability distributions; Statistical methods; Acoustic properties; Body waves; Seismic attenuation.

1 INTRODUCTION: THE NEED FOR A NEW MODEL PARADIGM

The earliest (1940s and 1950s) nuclear weapons tests were open-air, often higher-yield events. Event phenomenologies were monitored with different technologies. For example, tests were filmed using specialized cameras that captured the fireball expansion as a function of time, providing estimated yields. Optical sensors monitored the characteristic 'double hump' light flashes, which also contained yield information. Acoustic instrumentation monitored near-ground-level detonations by evaluating barometric pressure changes induced by explosions' blast waves. For other near-ground-level tests, surface effects were measured, for example, the dimensions of craters that were formed upon detonation.

With the Partial Test Ban Treaty in 1963, the U.S. and (then) Soviet Union ceased open-air nuclear testing due to environmental concerns. Their subsequent tests were conducted underground, rendering less valuable the optical, acoustic and surface effects approaches to monitoring. Seismometers detecting subtle ground motion then became the go-to remote monitoring technology. Yields of many early tests were publicly announced—see U.S. Dept. of Energy (2015) for a summary of U.S. tests and E.S. Vergino (1989) for a summary of some Soviet tests. If reported accurately, the announced yields were based on careful measurements and allow for calibration of seismic networks (R. Picard & M. Bryson 1992).

More recent world events have altered the nuclear forensics landscape. Clandestine activities in support of a weapons development program as well as nuclear terrorism are now significant concerns. In contrast to the Cold War era, which involved multiple high-yield events taking place at known test sites and allowed for calibration of seismic networks at great distances from those test sites, there is now substantial interest in potential lower-yield events that could be carried out at locations not known in advance and monitored by sensors that are unlikely to be calibrated to source-location-specific empirical explosion data.

In response to related concerns, controlled low-yield experiments involving high explosives have been conducted in recent years, some of which are described in later sections. In addition, there has been renewed interest in optical and surface effects data obtained during 1950s-era open-air nuclear tests that are relevant to certain potential terrorist nuclear incidents.

Yield estimation is an important initial step in the post-detonation forensics process for event reconstruction. As described by R. Stone (2016), the larger event reconstruction process (which includes the chemical analysis of radioactive debris and the running of complex computer simulations) can ultimately lead to identification of the perpetrator(s). Understanding the accuracy of the yield estimate informs this larger process.

To address yield estimation for smaller events at unknown test sites, the physical models valid for the Cold War era (large yield at known test sites) now have model error that must be captured in the ensemble error model for all monitoring analyses. This advanced model error accommodates both inadequate benchmark data and the mathematical structure of models.

In subsequent sections, we review the related issues, provide an improved modelling approach based on random effects methodology and an errors-in-variables treatment of yields used to benchmark the statistical models. Accounting for these considerations, the yield of a future event is inferred (with uncertainty) from each individual monitoring phenomenology. Finally, the multiphenomenology explosion monitoring (MultiPEM) methodology provides a unified yield estimate with uncertainty via data fusion across all individual phenomenologies. This approach is illustrated in an example.

2 STATISTICAL MODELLING ISSUES

2.1 Errors in variables

In the Cold War era, nuclear testing was carried out as noted. Yields of nuclear explosions were estimated in a variety of ways. Depending on the era that past events were conducted, publicly released yields, for example, U.S. Dept. of Energy (2015), were largely based on optical fireball or radiochemistry methods. More remote monitoring, such as involving seismic measurements, also provided estimates. Because of the inability to make error-free physical measurements, no approach allowed for error-free yield estimation.

Controlled experiments in recent years for forensics purposes involved conventional high explosives (HEs). These experiments were intended to help improve instrumentation to better characterize low-yield explosions.

Of current concern are the chemical-to-nuclear (C2N) conversion factors used to convert HE yields in those experiments to corresponding nuclear yields. Empirical results dating back to the early 1990s (M.D. Denny 1994) showed that several seismic signatures for fully coupled HE detonations behaved similarly to those from fully coupled nuclear events having roughly twice the explosive yield (controlling for geology). These results motivate modelling HE results as if they were nuclear results having ‘equivalent’ nuclear yields.

Non-nuclear materials used for HE events are subject to intrinsic manufacturing variability and conversion factors are not known exactly. Per C.E. Needham (2010, p. 55): ‘Unfortunately, there is no single method of establishing the equivalency of one explosive to another. Common methods currently in use include: pressure, impulse and energy equivalencies, each of which vary as a function of range.’ Many seismic/acoustic data analyses involve simplified, across the-board conversion factors, making it unrealistic to treat related equivalent yields as being truly known.

Importantly, error in the C2N conversion factor systematically affects *all* corresponding individual benchmark yields, affecting yield estimation for future events based on using those benchmark yields. More importantly, uncertainty in the conversion factor is usually ignored by traditional forensics models, which treat benchmark yields as being known exactly.

The uncertainty in an estimated yield for a future event is affected by uncertainties in the yield values used to calibrate predictive algorithms. If systematic error results from the use of a C2N conversion factor derived from one monitoring scenario (e.g. below ground, fully coupled events in a soft rock geology as monitored seismically) to another scenario having limited data (e.g. above ground events in a hard rock geology as monitored acoustically), uncertainties in yields of future events must account for the possibility. The errors-in-variables approach allows for formal uncertainty propagation of such systematic uncertainties. Other uncertainties that are associated with the individual sensor types also affect the weights guiding the process of obtaining the best overall yield estimate.

The example in Section 4 examines this issue in more detail.

2.2 Random effects and fixed effects

In the Cold War era, U.S. and Russian nuclear testing involved numerous events conducted at essentially the same locations, that is, the main test sites for the two countries. As such, the same source-to-sensor paths (‘fixed effects’) that were part of teleseismic monitoring were repeatedly observed, which allowed for the computing of individual station corrections whose purpose was to produce corrected metrics independent of sensor location and path. See, for example, S.R. Taylor & H.E. Hartse (1998) for a classical treatment.

Ideally, propagation path and station corrections could be made for future events. Unfortunately, future events may involve locations for which source-to-sensor paths and sensors themselves are observed only once (such as for a hypothetical terrorist incident) or perhaps 2 or 3 times, for example, in the initial stages of North Korean testing (D.B. Delbridge *et al.* 2023). Thus, there is not adequate empirical data for path-specific adjustments.

Even though the available source-to-sensor paths from benchmark experiments are not directly applicable to future events, for well designed benchmark experiments their source/path effects are statistically representative of those for future events. This allows future source and path effects to be regarded as samples from a statistical population and for characterizing variation among the benchmark experiments’ source/path effects that is useful in assessing uncertainties in future yield estimates.

The basic modelling concept is to replace a fixed-value station correction with a random variable that behaves according to the distribution of station corrections from the benchmark data. Intuitively, this replacement increases uncertainty in yield estimation relative to what would have been obtained ignoring source/path effects.

Table 1. Surface effects data for near-surface nuclear tests.

Test location	Sugar NTS	Cactus Eniwetok	LaCrosse Eniwetok	Koon Bikini	Zuni Bikini	Mike Eniwetok	Bravo Bikini
Yield (kt)	1.2	18	39.5	110	3530	10 400	15 000
Height of burst (ft)	3.5	3	17	13.6	9	35	15.5
Crater radius (ft)	45	185	202	400	1240	2800	3000
Crater depth (ft)	21	36	44	40	115	164	240

It is reasonable to assume that the average expected source/path effect is zero for the unknown ('random effect') source/path—zero being the average of fixed-effects adjustments in hypothetical benchmark data, and no reason that the relevant adjustment for a future event is more likely in one direction (positive) than the other (negative). In addition, substituting 'zero' for a source/path effect corresponds to the forward model prediction from an event. Clearly, the 'zero' has an associated uncertainty related to source/path variation that requires inclusion in the yield uncertainty.

2.3 Yield extrapolation

Under the Cold War monitoring mission, models based on first-principles physics were used for known test sites and needed parameter calibration. Monitoring tests at lower yields becomes problematic as the monitoring problem differs from that under a Cold War environment. In statistical terms, considerable care is needed when extrapolating from HE benchmark yields, which may be orders-of-magnitude lower than yields for future events. Similarly, yields for past nuclear events (e.g. in Table 1), may be much larger than yields for future events, requiring extrapolation in the other direction. Height/depth of burst may also be extrapolated considerably for some forensic data sets.

2.4 Bayesian methods

The modelling herein is compatible with either traditional or Bayesian statistical methods. Past work in seismic monitoring, such as R. Picard & M. Bryson (1992) and R.H. Shumway (2001), has explored the Bayesian approach. Bayesian methods are most practical when (a) expert information is available, (b) it agrees with reality, that is, to within presumed uncertainties and (c) it is free of controversy, for example, controversy when different experts disagree on the merits of monitoring phenomenologies or on the relative likelihoods of event scenarios.

Considerable care must be used in application of Bayes methods in nuclear forensics work. Errors-in-variables models for MultiPEM applications can involve high-dimensional parameter spaces, which complicate the assessment of Markov chain Monte Carlo convergence. Further, non-informative prior distributions should usually be avoided: Best case they provide results almost identical to traditional analysis, and worst case not only do such priors misrepresent whatever expert knowledge does exist (an important consideration for subjectivist Bayesians), but they can produce misleading results for nonlinear forward models used in forensics because flat priors in parameter space *do not* imply flat priors on corresponding nonlinear predictive surfaces (J.W. Seaman *et al.* 2012). These cautions are not intended to preclude the application of Bayesian data analytic techniques, but instead to ensure that all-too-common pitfalls of 'cookbook' Bayesian applications are avoided when applied to the forward models common to nuclear forensic data.

3 MONITORING PHENOMENOLOGIES

The four monitoring phenomenologies examined—surface effects, optical, seismic and acoustic—are presented solely for illustration. Similarly, the eight corresponding sensor metrics (two crater dimensions for surface effects, the times $t_{\min}$ and $t^{(2)}_{\max}$ for optical, *P*-wave displacement and maximum ground velocity for seismic, overpressure impulse and duration for acoustic) are discussed. For each sensor metric, a specific forward model is described. There is no claim that these specific metrics and models are the only such useful choices, or that other choices could not be considered.

The major points are that (a) all relevant data sources for an event should be used for yield estimation and (b) the new modelling paradigm is compatible with all forensic metrics and forward models that we are aware of.

3.1 Geophysical surface effects

For a nuclear device detonated above ground and at sufficiently low heights, a crater is formed upon explosion by an acoustic shock front on the earth. Crater radius (more formally defined as the radius of a best-fitting circle to the crater lip position data) and crater depth are log-linearly related to the yield of the explosion owing to various physics considerations (S. Stein & M. Wysession 2013). See Fig. 1.

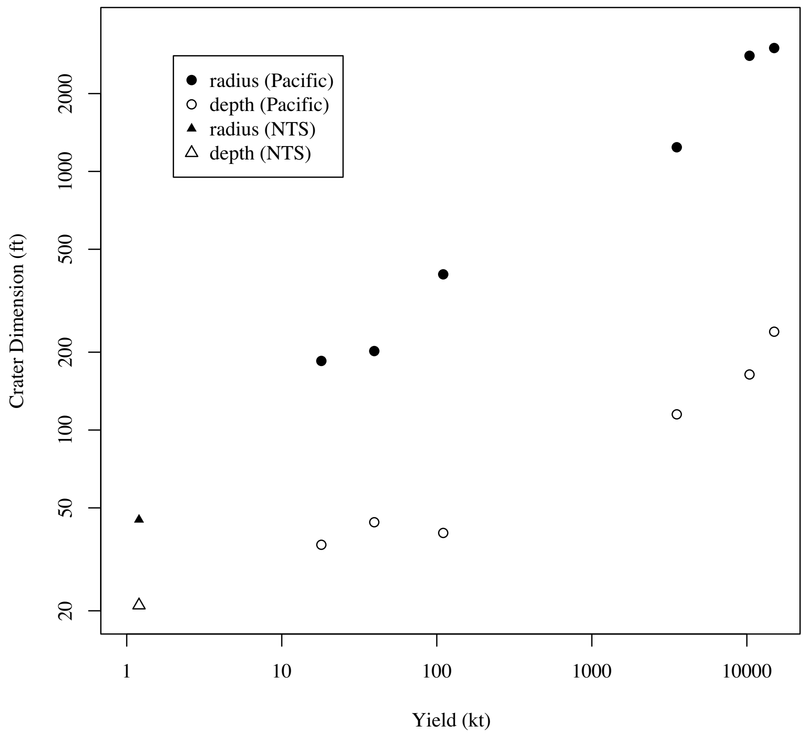

Figure 1. Geophysical surface effects data for near-surface nuclear explosions.

Surface effects data transcribed directly from L.J. Circeo & M.D. Nordyke (1964) are given in Table 1 for several 1950s-era nuclear tests conducted above ground. Shortly thereafter, the U.S. ceased open-air nuclear testing, precluding availability of additional data for this type of event. Data taken from the events in Table 1 are plotted in Fig. 1.

For completeness, if the height-of-burst (HOB) greatly exceeded those in the data, crater dimensions would diminish, although there is no indication of a HOB effect within the limited HOBs in Table 1. Further, were the new event to occur below the ground surface, subsidence and/or ejecta craters would be formed when the earth collapses into cavities formed by the detonations and/or is ejected by the blast. The physical mechanism for such cratering is more complex for below-ground tests than for tests slightly above ground, although crater dimensions still behave as functions of yield and geology (G.H. Higgins 1970). Such modestly below-ground events are not relevant to the two most common scenarios of greatest forensics interest, namely (a) maximizing destruction by not burying the detonation at all and (b) evading detection by burying the detonation much more deeply.

3.2 Optical

Optical methods were used in conjunction with the earliest nuclear tests. When a nuclear device is detonated above ground, a strong light flash is emitted. A special ('streak') camera allows for very fine time resolution to capture the pixel intensity of the light flash as a function of time. Recent application of modern image processing techniques to the original films (H. Auten 2017) has led to more precise optical data than was previously available.

Fig. 2 illustrates the time dependence of the pixel intensity for the 1950s-era nuclear test Sugar. Note the 'double hump' optical signature, where the first hump is related to the initial shock front and the second hump is related to thermal radiation. This double-hump shape is characteristic of nuclear explosions (G.E. Barasch 1979), and related data from optical sensors placed on satellites have been an essential component in detecting clandestine above-ground nuclear events (W.B. Scott 1997).

From empirical observations and physical basis, two specific times—t_{min}, the time of the minimum trough between the two humps and $t^{(2)}_{\mathrm{max}}$, the time of the maximum of the second hump—are related to yield. This is apparent in Fig. 3, which includes open literature data in Table 1 from seven 1950s-era near surface nuclear tests—six of which were conducted in the Pacific and the other at the Nevada Test Site—as well as 20 additional benchmark tests, with all optical data provided courtesy of S.R. Ford *et al.* (2021).

For above-surface detonations, energy is reflected off the ground surface into the fireball, enhancing optical signals such as t_{min} and creating an appearance of increased yield. That is, in contrast to the surface effects data, there is a HOB effect to be accounted for in the optical forward model. Several forward models have been developed for this purpose; we consider a model from R. Whitaker & E. Symbalisty (2009).

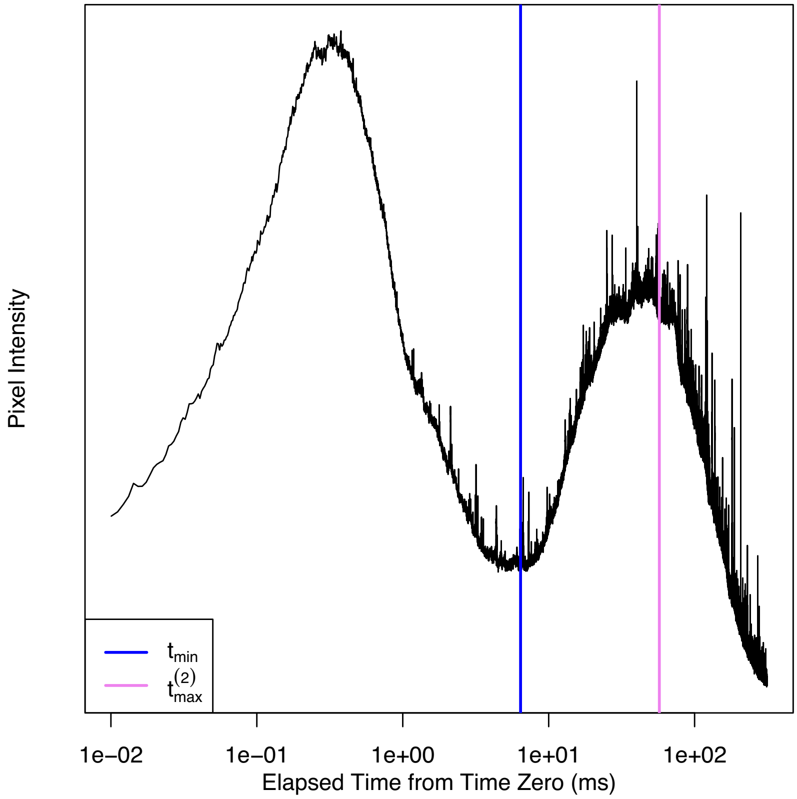

Figure 2. Light flash for the nuclear explosion Sugar.

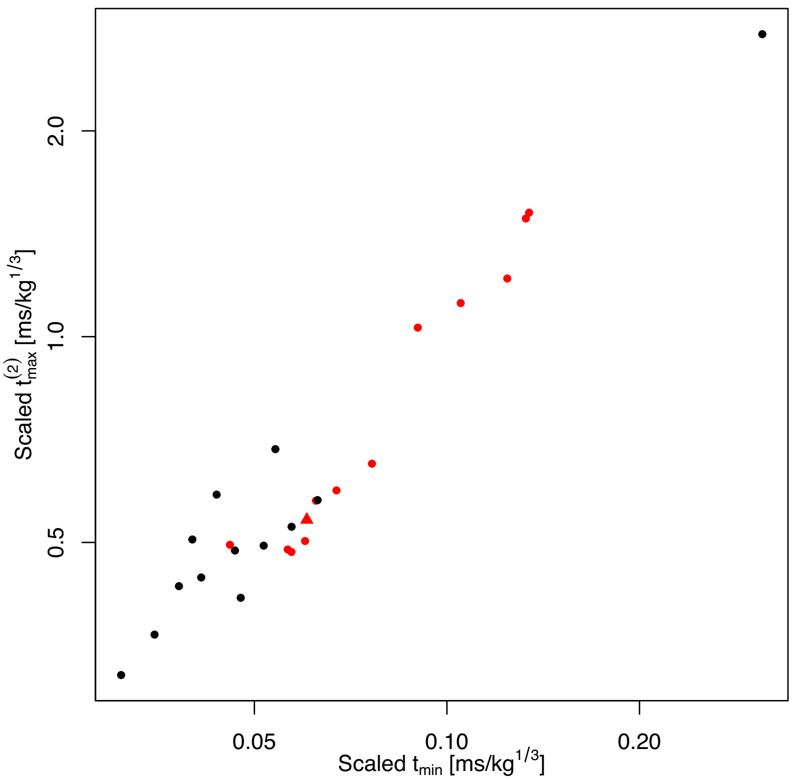

Figure 3. Optical data from S.R. Ford *et al.* (2021) for near-surface events. Benchmark (solid circles) and new event (solid triangle) data are shown. Colored symbols represent strongly surface-interacting events whose effective yield due to ground surface reflection is at least twice as large as airburst yield. Times and HOB are scaled by $w^{-1/3}$.

Table 2. Soft rock, above-ground HE benchmark data.

Experiment	HRI 1	HRI 2	HRI 3	HRI 5	HRIII 1	HTA 1	HTA 2	HTA 3
Yield (kg)	539.77	539.77	539.77	539.77	90.72	209.02	209.02	209.02
HOB (ft)	16.4	9.8	4.9	1.6	6.6	15.7	8.5	2.0
# seismic obs.	7	10	11	11	26	13	12	11
# acoustic obs.	2	3	2	3	53	14	14	18

The model begins with the usual log–log model for airburst events whose scaled HOBs are great enough to be free of ground reflection effects (using hydrodynamic and radiation transport simulation codes, they estimate this scaled HOB to be roughly 80 ft/kt$^{1/3}$),

$$\log t_{\text{min}}^{\text{airburst}} = \beta_0 + \beta_1 \log w \,,$$

where w denotes the explosion's nuclear yield and 'log' denotes the natural logarithm. Then, again using calculations from simulation codes, they arrive at the HOB smoothing function

$$\frac{t_{\text{min}}}{t_{\text{min}}^{\text{airburst}}} = 1 + \beta_2 \exp\left(-\left(\frac{h}{\beta_3 w^{1/3}}\right)^2\right),$$

where h denotes the height of burst. Combining these two relations produces the forward model

$$\log t_{\text{min}} = \beta_0 + \beta_1 \log w + \log\left(1 + \beta_2 \exp\left(-\left(\frac{h}{\beta_3 w^{1/3}}\right)^2\right)\right),$$

which provides a HOB adjustment to the usual log-linear model. A forward model of the same form (but with different β parameter values) is used for the $t_{\text{max}}^{(2)}$ data as well.

3.3 Seismic

Seismometers can provide useful yield data for low-yield nuclear explosions when located sufficiently close to the detonation. Several factors affect the ground motion recorded by a seismometer, broadly categorized into source effects (e.g. actual event yield, the extent of energy coupling with the ground) and source-to-sensor path effects (e.g. the type(s) of geology along the path, water content of the rock, signal attenuation and any sensor-specific biases that may exist).

Controlled experiments involving low-yield HE explosions and relatively closely placed seismometers have been conducted in an effort to better understand the issues involved. Metadata from eight such above-ground experiments, conducted in a soft rock geology, are provided by S.R. Ford *et al.* (2021) in Table 2 and displayed in Fig. 4. HOBs for these experiments ranged from 1.6 to 16.4 ft, and HE yields ranged between 90 and 540 kg.

Each series of experiments (Humble Redwood 'HR' and Humming Tarantula 'HTA') entailed multiple tests from the same source location as monitored by the same set of seismometers. As such, the situation is a small-area (within a few square miles) microcosm of teleseismic monitoring, and station corrections (source-to-sensor 'path effects') can be estimated.

Corresponding modelling is similar to that for a randomized block design in basic statistical textbooks (G.E.P. Box *et al.* 2005), whose purpose is to isolate and identify block effects and treatment effects. That is, observed sensor responses in a MultiPEM setting are modelled as

$$\begin{aligned}\text{sensor response} &= \text{mean value} + \text{block effect} + \text{treatment effect} + \text{model error}\\ &= \text{forward model} + \text{source effect} + \text{source-to-sensor path effect} + \text{model error}\,,\end{aligned}$$

which can be rearranged to give

$$\text{sensor response} - \text{forward model} = \text{source effect} + \text{source-to-sensor path effect} + \text{model error}\,. \qquad (1)$$

Were the forward model to be perfect in every way, containing all relevant factors and incorporating them correctly, then no unmodelled source or path effects would exist. Because no forward model is ever perfect, having methods to quantify the sizes of those unmodelled effects is important.

The randomized block structure allows for a simple, familiar process to check for unmodelled source/path effects, especially for forward models that explicitly incorporate effects that are source-specific, such as height of burst (which affects energy coupling) or path-specific, such as source-to-sensor distances (which affects signal attenuation). Per the randomized block design concept, the observed model errors on the left hand side of (1) from the forward-model-only fit can be arranged in a two-way layout, with rows defined by the different sources and columns defined by the different paths. This possibly unbalanced two-way layout can then be analysed using the standard analysis of variance table to test for the significance of unmodelled source and/or path effects. If significant effects are observed, the specific sources and/or specific paths that are consistently overestimated (or consistently underestimated) by

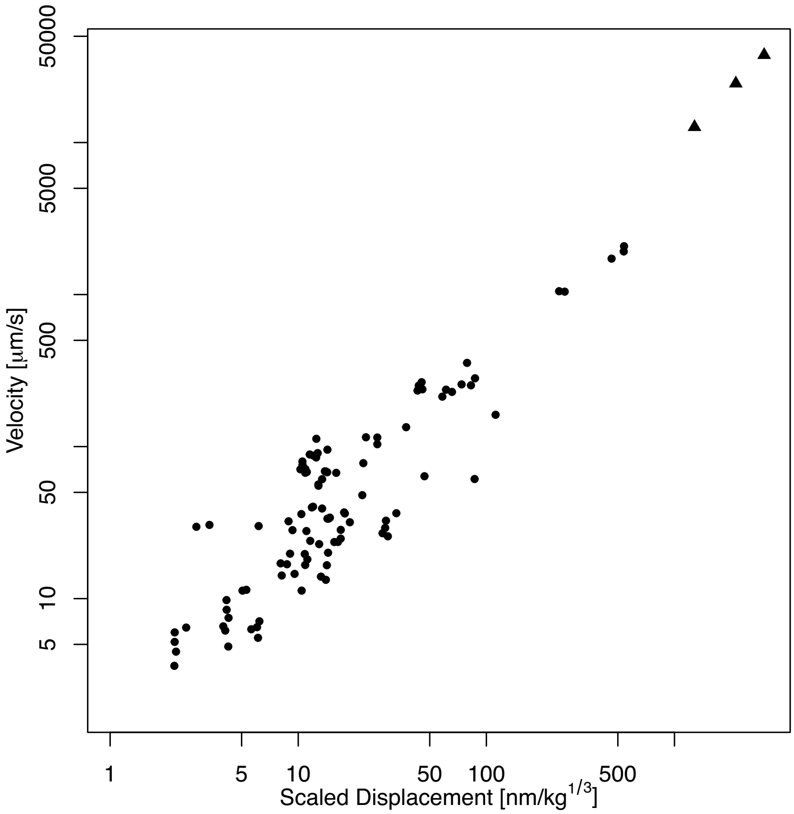

Figure 4. Seismic data from S.R. Ford *et al.* (2021) for near-surface events and soft rock geology. Benchmark (solid circles) and Sugar's nuclear data with C2N $= 2$ (solid triangles) are shown. Displacement is scaled by $w_{\mathrm{HE}}^{-1/3}$.

the forward model can be identified. Such knowledge of these unmodelled effects is then incorporated into uncertainties in estimates made for future events.

For illustration, we use forward models taken from S.R. Ford *et al.* (2021) that rely heavily on a postulated cube root scaling. Two important metrics, namely the *P*-wave maximum 3-D ground velocity and the displacement (equal to the integral of velocity over the *P*-wave time interval), are used in yield estimation. The forward models for these metrics have the general form

$$\log \text{response} = \beta_1 + \beta_2 \log \tilde{r} + \beta_3 \,\text{logistic}(\beta_4 + \beta_5 \tilde{h})\,, \tag{2}$$

where

(i) 'response' denotes either velocity or scaled (by $w_{\mathrm{HE}}^{-1/3}$)- displacement,
(ii) $\tilde{r}$ is the scaled source-to-sensor distance, $\tilde{r} = r/w_{\mathrm{HE}}^{1/3}$,
(iii) $\tilde{h}$ is the scaled height of burst, $\tilde{h} = h/w_{\mathrm{HE}}^{1/3}$,
(iv) the logistic function is $\text{logistic}(x) = 1/(1 + \exp(-x))$,
(v) the β's are response-specific model parameters, and
(vi) w_{HE} is the HE explosion yield.

Combining the forward model (2) with the source/path effects and model error (1) gives the full model

$$\log \text{response} = \beta_1 + \beta_2 \log \tilde{r} + \beta_3 \,\text{logistic}(\beta_4 + \beta_5 \tilde{h}) + S + P + e_{\mathrm{r}}\,, \tag{3}$$

where the S denotes the random source effect for the specific sensor response, P denotes the random source-to-sensor path effect for the specific sensor response and e_{r} denotes the model error. Estimates for the S's and P's across the set of sources and source-to-sensor paths in the calibration data are then obtained. Typical of randomized block designs, the observed source and path effects are centred around zero, and from the central limit theorem (D. McAlister 1879) tend to be Gaussian distributed because they represent the accumulation of numerous smaller factors. Calibration data are used to characterize statistical populations $N(0, \sigma_S^2)$ for the unmodelled source effects and $N(0, \sigma_P^2)$ for the unmodelled source-to-sensor path effects. The case study in Section 4 gives an example of this process.

Upon rearranging the approximate proportional relation, velocity $\propto$ displacement/$w_{\mathrm{HE}}^{1/3}$ in Fig. 4, a strong proportionality exists between the HE yield w_{HE} and the cubed ratio of displacement to velocity. Second order source- and path-related terms, scaled HOB and scaled source-to-sensor distance, are also incorporated into forward modelling as well. See Section 4.4.

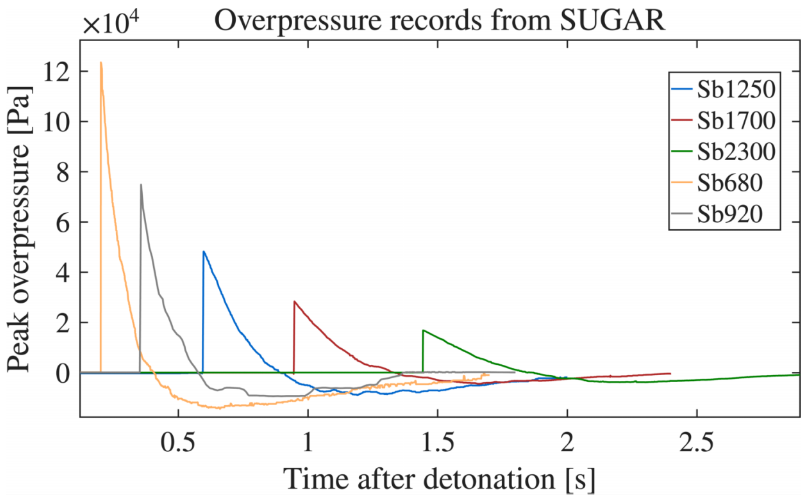

Figure 5. Acoustic waveform data for Sugar collected at five individual sensors. The source-to-sensor distance (ft) is appended to 'Sb'.

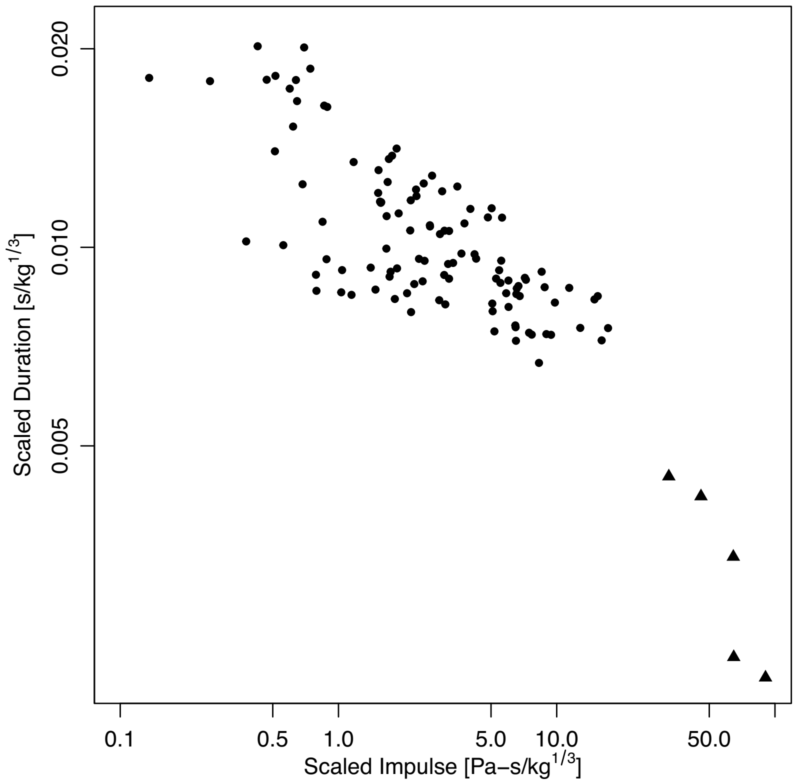

Figure 6. Acoustic data from S.R. Ford *et al.* (2021) for near-surface events and soft rock geology. Benchmark (solid circles) and Sugar's nuclear data with C2N = 2 (solid triangles) are shown. Impulse and duration are scaled by normalized temperature, normalized pressure, and $w_{\mathrm{HE}}^{-1/3}$.

3.4 Acoustic

The shock front from an above-ground explosion compresses the air immediately ahead of it. As the shock front passes a sensor location, this creates a spike in barometric pressure. See Fig. 5 for an illustration based on acoustic data from several sensors used for the nuclear test Sugar.

Two characteristics of the pressure change are the overpressure impulse and duration. The acoustic impulse is the numerical integral of the overpressure curve between the time of the shock front's arrival until the pressure first returns to ambient and is thus similar to the seismic *P*-wave displacement. The acoustic duration is simply the elapsed time between the shock front's arrival and the first return to ambient. Acoustic data from the same eight above ground experiments of Section 3.3 in Table 2 are provided by S.R. Ford *et al.* (2021) and displayed in Fig. 6.

Forward models proposed by S.R. Ford *et al.* (2021) for the impulse and duration begin by computing each calibration experiment's normalized temperature and pressure by dividing the ambient temperature and pressure at the time of the experiment by the standard temperature (288° K) and pressure (101 325 Pa) at sea level, respectively. The scaled impulse and duration are computed by dividing the observed impulse and duration by $w_{\mathrm{HE}}^{1/3}$ and then multiplying by the normalized temperature and normalized pressure accordingly.

Similar to the seismic forward models, the acoustic forward models have the generic form

$$\log \text{response} = \beta_1 + \beta_2 \log \tilde{r} + \log \text{logistic}(\beta_3 \tilde{h}),$$

where

(i) 'response' denotes either scaled impulse or scaled duration,
(ii) $\tilde{r}$ is the scaled source-to-sensor distance, $r/w_{\mathrm{HE}}^{1/3}$,
(iii) $\tilde{h}$ is the scaled height of burst, $h\,/\,w_{\mathrm{HE}}^{1/3}$,
(iv) the logistic function is $\text{logistic}(x) = 1/(1 + \exp(-x))$, and
(v) the β's are response-specific model parameters.

[Aside: Knowledge of the ambient temperature and pressure as well as the source-to-sensor distance, as needed by the forward models and which are known for the calibration data, could be problematic for certain future events, which can limit the usefulness of these particular forward models in practice.]

3.5 Yield modelling

Yield uncertainty is addressed by modelling yields for benchmark nuclear tests as being measured up to a user-specified error. Specifically, a common errors-in-variables treatment here provides for an additive error in log yield scale, that is,

$$\log w_{\mathrm{m}} \sim \log w + e_{\mathrm{w}}\,, \quad e_{\mathrm{w}} \sim \mathcal{N}(0, \sigma_{\mathrm{w}}^2)\,,$$

where w_{m} is the measured yield and w is the actual (unknown) yield. The standard deviation parameter σ_{w} is specified by the user, and controls the degree of deviation e_{w} that is expected between the actual and measured yields. Yield measurement errors for different events are considered to be statistically independent with common variance.

Analysis of HE benchmark data produces high-explosive yield estimates $\widehat{w}_{\mathrm{HE}}$. Statistical inference is typically done in log scale, leading to the Gaussian paradigm

$$\log \widehat{w}_{\mathrm{HE}} \sim \mathcal{N}(\log w_{\mathrm{HE}}, \sigma_{\mathrm{HE}}^2)\,,$$

where σ_{HE} denotes model uncertainty involving the benchmark data.

When using data from controlled experiments involving high explosives to infer the nuclear yield of a future event, a somewhat more complex uncertainty characterization is needed, including an error in the C2N yield conversion factor. Early work (M.D. Denny 1994) related to fully coupled underground events indicated that an approximate factor C2N ≈ 2 appeared to be reasonable for certain seismic metrics.

Uncertainties in this factor arise from several sources. Overlaying and comparing signatures such as seismic *P*-waves from HE and nuclear events necessarily involves noisy data. It also may be necessary to extrapolate results from one set of circumstances having modest historical data (e.g. fully coupled below ground device emplacement in one geology, as monitored seismically) to another set of circumstances having little/no historical data (e.g. partially coupled above ground device emplacement in a different geology, as monitored acoustically). The subjective modelling assumptions guiding such extrapolation also introduce uncertainty—uncertainty that is almost always underestimated by standard statistical methods (C. Tong 2019). Moreover, C2N uncertainty introduces correlations in yield estimates for future events (e.g. correlating the seismic-only and acoustic-only yield estimates in the example of Section 4 that are based on the same C2N factor).

Whatever data-based estimation approach is used, the end result is an estimated C2N factor, denoted $\widehat{\mathrm{C2N}}$. In (the more Gaussian) log scale, the corresponding estimated uncertainty is denoted σ_{C2N}, or

$$\log \widehat{\mathrm{C2N}} \sim N(\log \mathrm{C2N}, \sigma_{\mathrm{C2N}}^2)\,.$$

When a one-size-fits-all factor such as $\widehat{\mathrm{C2N}} \approx 2$ is used across a broad array of circumstances and there is little available direct data on a particular circumstance of interest, the uncertainty in $\widehat{\mathrm{C2N}}$ can matter.

Estimation of the yield conversion factor C2N for a specific scenario (e.g. slightly above ground detonation, soft rock geology, and relevant to a specific sensor metric) would constitute a full article on its own. Uncertainties in an estimated conversion factor can be substantial, especially in light of limited supporting nuclear data for certain scenarios and the corresponding need to use modelling assumptions to leverage/extrapolate C2N-like results from a scenario having historical event data to another scenario without such data.

Converting from an estimated HE yield $\widehat{w}_{\mathrm{HE}}$ to its corresponding estimated nuclear-equivalent yield $\widehat{w}$ is done multiplicatively, that is,

$$\widehat{w} = \widehat{\mathrm{C2N}} \times \widehat{w}_{\mathrm{HE}} \Leftrightarrow \log \hat{w} = \log \widehat{\mathrm{C2N}} + \log \widehat{w}_{\mathrm{HE}} ,$$

from which it follows that

$$\log \widehat{w} \sim N \left(\log \mathrm{C2N} + \log w_{\mathrm{HE}}, \sigma^2_{\mathrm{C2N}} + \sigma^2_{\mathrm{HE}} \right) .$$

In cases where σ_{C2N} is not small relative to σ_{HE}, accounting for C2N uncertainty can make a difference. An example is provided in the next section.

4 CASE STUDY

4.1 A hypothetical nuclear terrorism scenario

The example scenario is straightforward. Terrorists acquire and detonate a nuclear bomb in a high-visibility location, and the resulting explosion causes considerable loss of life, significant destruction and widespread radioactive contamination. In the absence of a genuine nuclear terrorist detonation, consider using open literature data from the 1951 nuclear test Sugar. Sugar was a low-yield device (1.2 kt) that was detonated near ground level (height of burst 3.5 ft) at the Nevada Test Site. Forensic data for this example are taken to be the actual measurements from the Sugar experiment (see Tables 1 and 2).

Determining the height of burst based on available sensor data is problematic here. This is important because all signals are affected by burst height, the optical and acoustic signals strongly so. The observed crater characteristics together with the lack of an apparent delivery mechanism are incompatible with a classic airburst. Nonetheless there is a (limited) range of HOBs that can not be ruled out, and for which the available sensor data alone is incapable of narrowing. In what follows, yield estimates are obtained for a set of HOB values to quantify more exactly the sensitivity to HOB within the range of credible values. Four HOB values—0 ft (ground level), 3.5 ft (Sugar's actual HOB), 25 ft (a conservative upper limit for HOBs in the terrorism scenario), and 100 ft (simply to stress the models)—are considered to span a range of HOBs.

4.2 Geophysical surface effects

The simplest and most straightforward modelling involves the surface effects data because there are no path effects and HOB effects are negligible compared with model noise over the small range of HOB values for the scenario. As might be surmized from Fig. 1, the forward model postulates a linear relation in log–log scale between each sensor response type (crater radius and depth) and yield.

That is, for each specific surface effects benchmark event,

$$\begin{aligned} \log \text{response} &= \beta_0 + \beta_1 \log w + e_{\mathrm{r}} \quad \text{and} \\ \log w_{\mathrm{m}} &= \log w + e_{\mathrm{w}} , \end{aligned} \tag{4}$$

where w_{m} denotes the measured yield for the event, w denotes the event's actual yield and (β_0, β_1) are the response-type-specific intercept and slope parameters for the linear relation. The yield measurement error e_{w} for the 1950s-era nuclear test yields is modest, taken here to have a relative standard deviation of 10 per cent. The model error $e_{\mathrm{r}} \sim N(0, \sigma_{\mathrm{r}}^2)$, where the response-type specific standard deviation σ_{r} is estimated from the benchmark data.

Obtaining the surface effects yield estimate for this example involves combining 18 model equations of the form (4) for the six benchmark events (each event corresponding to a measured yield, crater radius and crater depth) with the two model equations from the event (Sugar) of interest for measured crater radius and depth. [Note: it is obviously not legitimate in this example to use Sugar's measured yield $w_{\mathrm{m}} = 1.2$ kt to 'predict' Sugar's actual yield w.]

In vector notation, letting $\mathbf{x}$ denote the 20 data points (i.e. seven measured crater radii, seven measured crater depths and six measured yields for the benchmark events, the overarching Gaussian nonlinear model can be written

$$\mathbf{x} \sim N\left(\mathbf{f}(\theta), \boldsymbol{\Sigma}(\theta)\right) \tag{5}$$

for θ the vector of 14 unknown parameters. That is, θ consists of the $(\beta_0, \beta_1, \sigma_{\mathrm{e}})$ for crater radius, the $(\beta_0, \beta_1, \sigma_{\mathrm{e}})$ for crater depth, the correlation between the logarithms of crater radius and depth, the six actual yields for the benchmark data and, most importantly, the actual yield for the event of interest. In (5), the vector $\mathbf{f}(\theta)$ consists of the 20 forward model equations that follow from (4), and $\boldsymbol{\Sigma}(\theta)$ denotes the covariance structure for $\mathbf{x}$.

The general modelling format in (5) is consistent with established models and the physical basis of waveform signal generation for many phenomenologies once the specific forward models embedded in $\mathbf{f}(\cdot)$ and the response covariance matrix $\boldsymbol{\Sigma}(\cdot)$ are properly defined. The Gaussian model structure (5) is also very common in the statistics literature. Under general conditions (B. Efron & D.V.

Table 3. Maximum likelihood estimates of variance components σ_S, σ_P and σ_r. For seismic and acoustic respectively, S1 is log-scaled displacement and log-scaled impulse and S2 is log velocity and log-scaled duration.

Phenomenology	σ_S		σ_P		σ_r	
	S1	S2	S1	S2	S1	S2
Seismic	0.20	0.32	0.37	0.39	0.29	0.43
Acoustic	0.40	0.18	0.14	0.11	0.27	0.13

Hinkley 1978; B. Efron 1982), the maximum likelihood estimate $\widehat{\theta}$ behaves as

$$\widehat{\theta} \sim N\left(\theta, [\mathbf{I}(\theta)]^{-1}\right) , \tag{6}$$

where $\mathbf{I}(\theta)$ is the Fisher information matrix, whose $(p, q)^{\text{th}}$ element of the information matrix is (L. Malago & G. Pistone 2015)

$$\mathbf{I}(\theta)_{p,q} = \left[\frac{\partial \mathbf{f}(\theta)}{\partial \theta_p}\right]^T [\boldsymbol{\Sigma}(\theta)]^{-1} \left[\frac{\partial \mathbf{f}(\theta)}{\partial \theta_q}\right] + \frac{1}{2} tr \left\{ \frac{\partial \boldsymbol{\Sigma}(\theta)}{\partial \theta_p} [\boldsymbol{\Sigma}(\theta)]^{-1} \frac{\partial \boldsymbol{\Sigma}(\theta)}{\partial \theta_q} [\boldsymbol{\Sigma}(\theta)]^{-1} \right\} .$$

Computing the maximum likelihood estimate $\widehat{\theta}$ and the observed information matrix $\mathbf{I}(\widehat{\theta})$ for the surface effects data, extracting the component of $\widehat{\theta}$ from (6) for Sugar's estimated log yield and combining with its corresponding variance in $[\mathbf{I}(\widehat{\theta})]^{-1}$ gives

$$\log \hat{w} \sim N(\log 1.127, 0.367^2) .$$

Upon exponentiating the usual 'plus or minus two sigma' 95 per cent confidence interval for the Sugar's log yield, the confidence interval in yield scale is

$$\exp\left(\log 1.127 \pm 2 \times 0.367\right) = (0.54\ \text{kt}, 2.35\ \text{kt}) ,$$

giving roughly a multiplicative factor-of-2 quality estimate in spite of the considerable extrapolation in yield from the benchmark data (whose minimum yield over the six benchmark events was 18 kt).

4.3 Optical

Recall from Section 3.2 that the forward models for $t_{\min}$ and $t^{(2)}_{\max}$ involve yield and height of burst. For a postulated HOB of 3.5 ft, the benchmark data provide the optical-only yield estimate (0.88 kt) and 95 per cent CI (0.17 kt, 4.61 kt). The roughly factor-of-5 confidence interval width is largely attributed to uncertainties related to the energy ground reflection effect at low HOBs. Consequently, the joint surface effects and optical ('SE + O') estimate heavily emphasizes the surface effects data relative to the optical data. That joint estimate is $\hat{w}_{\text{SE+O}} = 1.08$ kt, with an associated 95 per cent confidence interval of (0.55 kt, 2.13 kt).

The optical and surface effects 95 per cent confidence interval is found similar to the optical only confidence interval. That is, augment the data vector '$\mathbf{x}$' containing only the optical measurements with the additional measurements from the surface effects data, augment the parameter vector 'θ' for the optical model only with the additional parameters from the surface effects data, augment the functional mean '$\mathbf{f}(\theta)$' for the optical data only with the additional functional components for the surface effects data, and derive the covariance matrix '$\boldsymbol{\Sigma}(\theta)$' for the optical data to account for the joint optical and surface effects data in the newly constructed '$\mathbf{x}$'. At that point, the general Gaussian theory for (5) can be applied.

4.4 Seismic, acoustic and the HE benchmark events

Results for the seismic and acoustic phenomenologies are slightly more complex. The forward models are derived using sensors having relatively short source-to-sensor distances (e.g. the three seismometers and five acoustic sensors for Sugar were all located within a half mile of ground zero). These models explicitly incorporate HOB and source-to-sensor distances.

HOB effects can be substantial in certain cases, and are analogous to well known depth-of-burial effects for the more familiar case of underground explosions. As an example, seismic data may well be incapable of distinguishing a lower-yield event at a shallow depth of burial from a somewhat higher-yield event at a somewhat greater depth of burial, for example, J.R. Murphy *et al.* (2010, p. 461). For this case study example, optical-only and (to lesser degree) acoustic-only yield estimates are sensitive to HOB effects.

In the above-ground nuclear terrorism scenario at hand, however, the location of the crater allows for 2-D identification of the 3-D detonation location. When this is coupled with a presumed HOB value to complete the point source location, the forward models can then be used. As noted in Section 4.1, the range of credible HOBs is limited in the nuclear terrorism example, so that (see Table 4 in the next section) surface effects and seismic yield estimates within this limited range are not overly sensitive to HOB.

The full seismic forward model (3) partially captures source effects through yield estimation error and partially captures path effects through explicit incorporation of HOB and source-to-sensor distance. The random effect model parameters σ_S and σ_P for unmodelled source and path effects would equal zero were the model to be perfect. Best estimates of these quantities for this example are provided in Table 3, and show that these effects are commensurate with the residual standard deviation σ_r from the model fit. As would be

Table 4. MultiPEM yield estimates (kt) and 95 per cent confidence intervals for the Sugar event.

Monitoring Phenomenology	Height of burst (ft) 0	3.5	25	100
Surface effects only	1.13	1.13	1.13	1.13
	(0.54, 2.35)	(0.54, 2.35)	(0.54, 2.35)	(0.54, 2.35)
Optical only	0.38	0.88	2.51	0.78
	(0.03, 4.24)	(0.17, 4.61)	(0.27, 23.21)	(0.15, 4.03)
Surface effects & optical	1.05	1.08	1.15	1.08
	(0.52, 2.13)	(0.55, 2.13)	(0.57, 2.32)	(0.55, 2.14)
seismic only	0.67	0.67	0.68	0.68
	(0.22, 2.09)	(0.22, 2.09)	(0.22, 2.12)	(0.22, 2.13)
Acoustic only	1.04	1.02	0.91	0.72
	(0.36, 2.96)	(0.36, 2.92)	(0.31, 2.69)	(0.24, 2.20)
Seismic & acoustic	0.80	0.79	0.76	0.70
	(0.35, 1.82)	(0.35, 1.81)	(0.33, 1.76)	(0.30, 1.63)
MultiPEM	0.94	0.95	0.97	0.91
	(0.55, 1.60)	(0.56, 1.61)	(0.57, 1.66)	(0.54, 1.55)

expected, unmodelled path effects for acoustic data, where paths travel through the atmosphere, are smaller relative to unmodelled source effects than is the case for seismic data, where paths travel through the earth.

The calculations for an HOB of 3.5 ft, the actual height of burst for Sugar, give a seismic-only HE yield estimate of 0.34 kt, with a 95 per cent confidence interval of (0.12 kt, 0.97 kt), and an acoustic-only HE yield estimate of 0.51 kt, with a 95 per cent confidence interval of (0.19 kt, 1.34 kt). The combined seismic and acoustic HE yield estimate $\hat{w}_{\mathrm{HE}}$ is 0.40 kt, with 95 per cent confidence interval (0.19 kt, 0.81 kt).

Converting HE yield estimates to their nuclear equivalents requires the use of a C2N conversion factor. M.D. Denny (1994) found that seismic signals from fully coupled underground HE tests were roughly comparable to those from nuclear events having roughly twice the nuclear yield. There is limited information in the open literature involving the effects of various factors on this presumed factor $\widehat{\mathrm{C2N}} = 2$. Such factors include the type of geology involved, above/below ground device emplacement, the phenomenology involved (e.g. seismic maximum *P*-wave velocity versus acoustic duration), explosion yield, etc.

Ideally, ample open literature data directly relevant to a low-yield above ground event on soft rock would exist to aid in estimation of the C2N factor for the Sugar example. As this is not the case, it is problematic to assign an uncertainty to the estimated $\widehat{\mathrm{C2N}}$, which is largely extrapolated from results under other circumstances.

For purposes of illustration, a relative standard deviation of 25 per cent is used in the following analysis to account for the one-size-fits-all application of the C2N factor. That is, the usual 95 per cent 'plus or minus two sigma' confidence interval in log scale corresponds to a multiplicative 'two sigma' factor of 1.5 in natural scale. For $\widehat{\mathrm{C2N}} = 2$, the corresponding multiplicative confidence interval in C2N scale is

$$\mathrm{C2N} \in (2/1.5,\, 2 \times 1.5) = (1.333,\, 3) = \exp\left(\log 2 \pm 2 \times 0.203\right) .$$

Accordingly, the Gaussian distribution for $\log \widehat{\mathrm{C2N}}$ is

$$\log \widehat{\mathrm{C2N}} \sim N\left(\log 2,\, 0.203^2\right) .$$

Were a different, and scenario specific, conversion factor other than $\widehat{\mathrm{C2N}} = 2$ or a different uncertainty to be warranted, obvious adjustments to the following results could easily be made.

Consider the joint seismic-acoustic results for the actual HOB = 3.5 ft. The estimated HE yield for Sugar here is

$$\log \hat{w}_{\mathrm{HE}} \sim N\left(\log 0.396\ \mathrm{kt},\, (0.359\ \mathrm{kt})^2\right) .$$

Combining with C2N to obtain the seismic-acoustic ('$S + A$') nuclear equivalent yield $\hat{w}_{\mathrm{S+A}}$ gives

$$\begin{aligned} \log \hat{w}_{\mathrm{S+A}} &= \log \widehat{\mathrm{C2N}} + \log \hat{w}_{\mathrm{HE}} \\ &\sim N\left((\log 2 + \log 0.396)\ \mathrm{kt},\, \left(\sqrt{0.203^2 + 0.359^2}\ \mathrm{kt}\right)^2\right) \\ &= N\left(\log 0.792\ \mathrm{kt},\, (0.412\ \mathrm{kt})^2\right) . \end{aligned}$$

The usual 95 per cent confidence interval for the nuclear equivalent yield is then

$$\exp\left(\log 0.792\ \mathrm{kt} \pm 2 \times 0.412\right) = (0.35\ \mathrm{kt},\, 1.81\ \mathrm{kt}) .$$

Interestingly, were uncertainties in $\widehat{\mathrm{C2N}}$ and in source/path effects all to have been completely ignored, the seismic-only nuclear equivalent yield for Sugar, based on the forward model alone and more than 100 seismic data points, is 0.46 kt, with a 95 per cent confidence interval of (0.20 kt, 1.06 kt), which fails to contain the measured Sugar yield 1.2 kt. In other words, confidence intervals that fail to account for all uncertainties can lead to misleading conclusions.

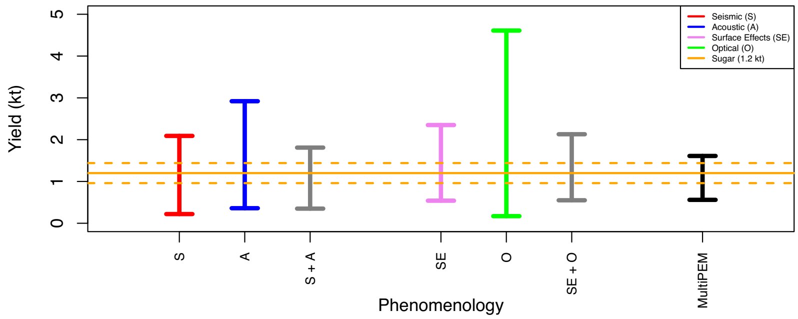

Figure 7. 95 per cent confidence intervals for estimated Sugar yield from the analyses in Table 4 corresponding to an assumed HOB = 3.5 ft, a 10 per cent relative standard deviation on benchmark nuclear yield estimates, 25 per cent relative error on the one-size-fits-all C2N = 2, and inclusion of random effects variances for unmodelled source/path effects.

4.5 Data fusion

MultiPEM is a paradigm for fusing signatures from multiple phenomenologies to improve characterization of new event device parameters with rigorous uncertainty quantification. In this subsection, its application is illustrated for yield characterization of Sugar using signatures collected from the surface effects, optical, seismic and acoustic phenomenologies as introduced in previous subsections.

Recall that the joint surface effects and optical yield estimate $\hat{w}_{\mathrm{SE+O}}$ based on the nuclear benchmark events was

$$\log \hat{w}_{\mathrm{SE+O}} \sim N(\log 1.079\ \mathrm{kt}, (0.339\ \mathrm{kt})^2)\,.$$

Combining with the joint seismic and acoustic yield estimate $\hat{w}_{\mathrm{S+A}}$ based on the HE benchmark events,

$$\log \hat{w}_{\mathrm{S+A}} \sim N(\log 0.792\ \mathrm{kt}, (0.412\ \mathrm{kt})^2)\,,$$

the usual weighted average (weighting inversely proportional to the respective variances) gives the MultiPEM yield estimate $\hat{w}_{\mathrm{MultiPEM}}$:

$$\begin{aligned}
\log \hat{w}_{\mathrm{MultiPEM}} &= \left(\frac{\sigma^2_{\mathrm{SE+O}}}{\sigma^2_{\mathrm{SE+O}} + \sigma^2_{\mathrm{S+A}}}\right) \log \hat{w}_{\mathrm{S+A}} + \left(\frac{\sigma^2_{\mathrm{S+A}}}{\sigma^2_{\mathrm{SE+O}} + \sigma^2_{\mathrm{S+A}}}\right) \log \hat{w}_{\mathrm{SE+O}} \qquad (7)\\
&\sim 0.404 \times N(\log 0.792\ \mathrm{kt}, (0.412\ \mathrm{kt})^2) + 0.596 \times N(\log 1.079\ \mathrm{kt}, (0.339\ \mathrm{kt})^2)\\
&= N\big((0.404 \times \log 0.792)\ \mathrm{kt} + (0.596 \times \log 1.079)\ \mathrm{kt},\\
&\qquad (0.404 \times 0.412\ \mathrm{kt})^2 + (0.596 \times 0.339\ \mathrm{kt})^2\big)\\
&= N(\log 0.952\ \mathrm{kt}, (0.262\ \mathrm{kt})^2)\,.
\end{aligned}$$

Thus, the MultiPEM yield estimate is 0.952 kt, and the corresponding 95 per cent confidence interval is

$$\exp(\log 0.952\ \mathrm{kt} \pm 2 \times 0.262) = (0.56\ \mathrm{kt}, 1.61\ \mathrm{kt})\,.$$

Other results for Sugar, assuming HOB values that are not the correct one but which could be plausible under the scenario, are given in Table 4. As seen in Fig. 7, joint analysis can lead to notable uncertainty reduction relative to single phenomenology analysis. This is particularly true when each individual phenomenology provides probabilistically consistent results for device parameters of interest, in this case yield. It is also noted that the MultiPEM model can characterize C2N and new event device parameters simultaneously when sufficient benchmark data exist to calibrate C2N, thereby avoiding the need for inverse-variance weighted conflation of distributions as in (7), though this is not the case for the Sugar example.

[Note. Full formulation of the MultiPEM model for the Sugar case study—with four monitoring technologies, eight sensor metrics, multiple seismic and acoustic sensors, corresponding multiple source/path effects per sensor, and all related model parameters—would entail several dozens of notational symbols and considerable matrix algebra, together with careful tracking of which sources/paths correspond to which observed metrics. Moreover, the complicated nature of the correlation structure in the covariance matrix $\mathbf{\Sigma}(\theta)$ is such that specialized software is needed for the calculations. In that such details are not the focus of the work, they are omitted in the interest of readability. B.J. Williams *et al.* (2025) reviews these calculations in detail, and the special case of linear forward models is treated in B.J. Williams *et al.* (2021) for the traditional setting in which benchmark yields are assumed to be known exactly. Also, theoretical and practical details regarding mixed effects models (of which the MultiPEM application is a special case) can be found in J.C. Pinheiro & D.M. Bates (2000).]

Major takeaways from this example include:

(i) The modelling paradigm presented here, involving explicit consideration of yield measurement errors in benchmark data, as well as a random effects treatment of transportable source/path variance components, is compatible with all forward yield estimation models. For current forensic applications, this approach provides more realistic uncertainty estimates than does traditional modelling, and also improves forward model parameter estimation by reducing the influence of larger source/path clusters.

(ii) The most practical impact of including (versus ignoring) uncertainties in the C2N factor and in random source/path effects is to widen confidence intervals. Estimated yields themselves are often less affected, but in a good way: the (modest) bias from traditional ordinary least squares estimation assuming 'known' yields is reduced as per errors-in-variables considerations (M.D. Levi 1973); such bias effects were magnified in the Sugar example because of the yield extrapolation from the very-low-yield HE results. In some cases, as with the seismic-only assessment of Sugar's nuclear yield, the more realistic confidence intervals provided by the new model help prevent data analysts from reaching mistaken conclusions.

(iii) The *MultiPEM Toolbox* (B.J. Williams 2024) is open source software developed in the *R* programming language at Los Alamos National Laboratory available to carry out this modelling (as well as to carry out traditional modelling).

(iv) MultiPEM yield estimates and confidence intervals for Sugar are robust to assumptions regarding the height of bursts relevant to the nuclear terrorism scenario. This is an important issue for future events because available sensor data alone may not allow for HOB to be accurately determined; along similar lines, typical forward models alone do not allow for accurate estimation of source-to-sensor distances, which in the example are determined using the location of the explosion's crater.

5 CONCLUSION

Advances in analysis to support nuclear explosion monitoring are constantly being developed. Data from open literature sources is used to test and validate these methods, thus providing confidence in their real-world application. We have developed and demonstrated the uncertainty propagation for a MultiPEM yield estimate. This advance explicitly incorporates uncertainties not previously accounted for. Most important among these are the source/path variation captured by the random effects modelling that is needed for potential future uncalibrated environments, as well as the uncertainty in the chemical-to-nuclear conversion factor used relative to ongoing controlled experiments with HE. As such, the model paradigm provides a more comprehensive error analysis for an all-encompassing explosion yield estimate.

Any future nuclear event would draw world-wide attention. Event characterization is a multistep process. Initial answers are obtained in a short (within a few hours) timeframe, and good uncertainty estimates are essential to communication of results. More thorough answers, from both open and official sources, would evolve over time by acquiring further information such as from chemical analyses of radioactive debris, complex computer simulations, post-event eyewitness input and various intelligence sources. Those more thorough answers can provide a more complete event reconstruction and aid in identifying the incident's perpetrator(s).

Statistical analysis of yield-related data from the event is an essential early part of this larger process, with a goal of fast turnaround time for reporting results. Consequently, benchmark data should be thoroughly analysed beforehand and that plans be made for scenarios that can be reasonably anticipated. Dry run exercises are also valuable.

Any conclusions that are publicly announced in the aftermath of an unexpected nuclear event would attract considerable scrutiny. Past history, for example, as involving the Vela incident (W.B. Scott 1997) and Cold War era nuclear weapons testing (R.J. Smith 1985), provides instructive examples of this scrutiny and of its residual effects, and highlights the importance of having defensible, well founded uncertainty estimates.

Relative to the yield estimation component, the modelling here is compatible with all forward models for sensor metrics that we are aware of. Although the point estimates of event yields from this new model paradigm are sometimes indistinguishable from those of contemporary methods, superior uncertainty estimation is achieved by replacing an amorphous model error term with components attributable to benchmark yield estimation uncertainty, transportable source and path effects and residual model error.

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DATA AVAILABILITY

The software and data sets (B.J. Williams 2025) used to benchmark forward model coefficients and estimate yield for each phenomenology are available by cloning the MultiPEM Toolbox GitHub repository (B.J. Williams 2024). Beginning from the root directory of the repository, the eight CSV formatted data files are located in the ./Applications/Data/IYDT-gsrp directory. The statistical analysis code is located in the ./Code directory, while input decks implementing the analyses supporting the results of this article are located in the ./Runfiles/IYDT-gsrp directory.

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